

Artificial Intelligence & Machine Learning – Task 3

Model Validation, Overfitting Control & Hyperparameter Tuning using Decision Tree Regression

Submitted By:

Rudra Dahikar

Introduction

Machine Learning enables computers to analyze data, identify patterns, and make predictions without explicit programming. Building an accurate model is important, but ensuring that it performs well on unseen data is equally essential. This project focuses on model validation, overfitting control, and hyperparameter tuning using a Decision Tree Regression model. The California Housing Dataset from Scikit-learn is used to predict median house prices based on various housing features.

Objective

The main objectives of this project are:

- To understand the California Housing Dataset.
- To perform Exploratory Data Analysis (EDA).
- To build a Decision Tree Regression model.
- To evaluate the model using RMSE, MAE, and R^2 Score.
- To apply Cross Validation for reliable performance evaluation.
- To optimize the model using GridSearchCV.
- To compare the performance before and after hyperparameter tuning.

Methodology

The project follows a structured Machine Learning workflow:

- Loaded the California Housing Dataset using Scikit-learn.
- Explored the dataset using descriptive statistics and information analysis.

- Performed data visualization using histograms, correlation heatmaps, boxplots, scatter plots, and feature importance graphs.
- Split the dataset into training and testing sets.
- Built a Decision Tree Regression model.
- Evaluated the model using MAE, RMSE, and R^2 Score.
- Applied 5-Fold Cross Validation to improve model reliability.
- Tuned hyperparameters using GridSearchCV to reduce overfitting and improve prediction accuracy.

Results

The Decision Tree Regression model successfully predicted housing prices based on the available features. Cross Validation helped evaluate the model more reliably, while GridSearchCV improved its performance by selecting the optimal hyperparameters. The tuned model achieved better generalization and reduced overfitting compared to the initial model. Feature importance analysis showed that median income and location-related features had the greatest impact on house price prediction.

Conclusion

This project successfully demonstrated the complete workflow of a Machine Learning regression problem, including data exploration, visualization, model building, validation, and optimization. The use of Cross Validation and GridSearchCV improved the overall performance of the Decision Tree model and reduced overfitting. This project strengthened my practical understanding of Machine Learning model evaluation, hyperparameter tuning, and predictive analytics using Python.

Tools & Technologies Used

- Python
- Jupyter Notebook
- NumPy
- Pandas
- Matplotlib
- Seaborn

- Scikit-learn

Skills Gained

- Exploratory Data Analysis (EDA)
- Data Visualization
- Decision Tree Regression
- Model Validation
- Cross Validation
- Hyperparameter Tuning
- GridSearchCV
- Model Evaluation using RMSE, MAE, and R^2 Score

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